

Project Nishu — Audio Pipeline & On-Device Model Report

Status summary | 3 October 2026

1. Objective and Architecture

Project Nishu V0.1 is an Android-first, local-processing assistant for recording/importing audio, transcribing speech, and generating summaries, tasks and decisions. The documented pipeline uses sherpa-onnx with an Oriserve Whisper-Hindi2Hinglish-Swift INT8 model for STT, followed by an on-device Qwen3-0.6B language model for text processing. The intended privacy benefit is that audio and transcripts can be processed without sending them to a cloud service.

2. Audio Pre-processing and VAD

VAD means Voice Activity Detection. Nishu uses Silero VAD to detect likely speech regions and help segment audio rather than sending long stretches of silence through STT. Reported settings include threshold 0.5, minimum speech 0.25 s, minimum silence around 0.25 s (some later notes say 0.45 s), and a maximum segment of 28 s. Because the notes differ on the silence setting and later batching limits, confirm the exact values in the current source code before treating them as final.

3. On-device Benchmark (reported test)

Test device: Motorola Edge 60 Fusion, MediaTek Dimensity 7400, Android 16, approximately 7.6 GB RAM. Test audio: a 42.7-minute, predominantly English product-marketing meeting.

4. Fine-tuned Qwen3-0.6B: Evaluation Snapshot

The supplied evaluation reports 71 tool-call examples and 150 normal-chat examples for the fine-tuned model. Tool-name accuracy was 71/71 (100%); exact argument accuracy was 64/71 (~90.1%); and false tool triggers were 1/150 (~0.7%). Calculator exact arguments were 12/16 (75%) and set_timer exact arguments were 7/10 (70%). These are results on a small test set, not proof of production readiness. A same-test-set comparison against stock Qwen3-0.6B has not been provided, so the improvement due to fine-tuning is not yet quantified.

5. Issues and Next Actions

- Run controlled STT ablations on the same audio: baseline, high-pass only, low-pass only, normalization only, and combined chain. Compare against a manually checked reference transcript (WER/CER), not word count alone.
- Verify the exact filter chain, VAD thresholds, segment batching and padding in the current Android source and installed build.
- Test English and Hinglish recordings separately. Evaluate language routing (English-oriented Whisper versus Swift) before adopting it.
- Review the seven tool-argument failures and the one false tool trigger; add held-out Hinglish and edge-case tests.
- Run stock and fine-tuned Qwen3-0.6B on identical prompts, decoding settings and scoring rules.
- Re-run the complete long-meeting workflow with the 36-chunk cap and confirm transcript, summary, memory, battery and thermal behavior on the physical device.
- Keep speaker diarization disabled or limited for long/noisy meetings until false speaker clusters are measured and controlled.

Status note: This report distinguishes reported implementation/test results from items that still need verification. “Build successful” and individual unit-test results do not, by themselves, establish transcription accuracy or production readiness.

Build and Verification Status

The app build is complete for the current debug build. The project notes report that gradlew :app:compileDebugKotlin completed with zero errors, gradlew testDebugUnitTest completed with zero test failures (27 tasks reported up-to-date), and gradlew :app:installDebug successfully installed the app on the Motorola Edge 60 Fusion. The notes also report that the redesigned UI was installed and running while the existing recording, STT, local LLM, Room database, and WorkManager pipeline were retained. These results confirm build and installation, but do not by themselves prove every long-audio or model-quality scenario works correctly.

Measured Tests and Results

Fixes Reported in the Current Build / Pipeline

- Replaced the English-only STT path for Hinglish recordings with the Whisper-Hindi2Hinglish-Swift INT8 model and updated annotation handling to avoid incorrectly showing “No speech detected.”
- Added Silero VAD to identify speech versus silence; reported settings include threshold 0.5, minimum speech 0.25 s, minimum silence 0.25 s (some later notes state 0.45 s), and maximum segment 28 s.
- Added 80 Hz high-pass filtering and level normalization (target around –20 dBFS, maximum gain boost +18 dB) to reduce low-frequency rumble and compensate for quiet audio.
- Reportedly removed aggressive pre-emphasis and disabled the intra-speech noise gate after they appeared to suppress useful speech cues and clip soft consonants/word onsets.
- Added short leading silence and speaker-boundary padding, plus transcript cleanup and email-formatting rules. Verify these details against the exact installed revision if a release audit is required.
- Moved long benchmark work to a foreground worker to avoid ANR; serialized heavy STT/LLM model use through EngineHolder to reduce concurrent memory pressure.
- Changed diarization behavior after long noisy recordings over-split speakers; long recordings now skip/cap diarization at 20 minutes.
- Raised the summary chunk cap from 12 to 36 and changed transcript clearing so existing text is not cleared until STT begins. These two changes were marked unverified in the test report because the final long-meeting rerun was interrupted.

Verification Caveats

- The report documents successful build, unit-test task completion and installation. It also explicitly says the final transcript/summary after the latest batching, transcript-clear-order and 36-chunk changes had not yet been verified on the physical phone.
- A complete regression run is still needed for the latest build: long meeting with 36 chunks, transcript quality, Room migration, memory gate in a separate process, screen-off recording, permission-denied UX, and real two-person Hinglish diarization.
- The Qwen3-0.6B fine-tuning evaluation is separate from the October 1 stock-model meeting benchmark. Do not imply the fine-tuned model produced that meeting summary unless the model was actually integrated and rerun on-device.

Component	What it does	Why it is used / status
80 Hz high-pass filter	Reduces very-low-frequency rumble and handling noise.	Documented in the pipeline notes; verify final device configuration.
7.5 kHz low-pass filter	Attenuates higher-frequency content above its cutoff.	Listed in the supplied notes; independent accuracy benefit has not been established.
Level normalization (~–20 dBFS target; up to +18 dB gain)	Brings recording level closer to a target level.	Documented target; may amplify background noise as well as speech.
Pre-emphasis (0.97)	Boosts higher-frequency components relative to lower ones.	Reportedly removed because it appeared to harm speech/formant cues and transcription.
Intra-speech noise gate	Suppresses audio below a threshold during speech.	Reportedly disabled because soft consonants and word onsets could be cut.
320 ms silence prefix	Adds a short silent lead-in to the audio.	Reported as a cold-start stabilization measure; verify on current build.
~200 ms speaker-boundary padding	Preserves audio around speaker-turn boundaries.	Reported as a mitigation for words being clipped at boundaries.
Email post-formatting	Converts spoken forms such as “name at the rate gmail dot com” to email syntax.	Reported as a post-processing rule; test on varied examples.

Metric	Reported result	Interpretation / caveat
End-to-end processing	860 s (~14.3 min)	Faster than real-time overall in this test.
STT	670 s; 778 segments	Transcript still had missed English words and short Hindi-like hallucinations.
Summarization	190 s	The first reported run covered only 12 chunks; later cap increased to 36, but the expanded run was not verified in the supplied report.
LLM prefill	168 tok/s, up from 47 tok/s	Reported CPU/runtime optimization; benchmark conditions should be kept consistent.
LLM decode	~20 tok/s	Reported on-device result.
LLM memory (PSS)	~642 MB, down from ~1.17 GB	Reported after disabling weight repacking; the stated 500 MB PSS target was not met.

Test	Measured result	What it tells us
Full offline meeting pipeline	42.7 min audio; 860 s (~14.3 min) total	Completed once end-to-end; faster than real time overall.
STT on meeting	670 s; 778 segments; ~6,245 words	Transcript contained missed English words and short Hindi-like fragments; quality remains an issue.
LLM summary / extraction	190 s total; 12 of 25 transcript chunks processed in that run	Summary was generic and covered only about the first 24 minutes. The later 36-chunk cap was added but its full meeting result was not verified in the report.
STT short recording	33.8 s audio processed in ~14.5 s with 4 CPU threads; ~18 s with 1 thread	CPU STT was faster than real time on this sample; more threads helped modestly.
LLM prefill	47 → ~168 tokens/s	About 3.6× improvement after selecting the ARMv8.2 dot-product/FP16 CPU runtime variant.
LLM decode	~15 → ~20 tokens/s	Measured speed improvement after runtime/thread tuning.
Prefix-cache warm load	6.3 s cold prefill → ~12 ms warm load	Reported for the same prefix; output was identical.
LLM memory	~1.17 GB → ~642–650 MB PSS	Disabling weight repacking and using ubatch 128 saved about 380 MB, but the 500 MB PSS target remains unmet.
NNAPI vs CPU for STT	68.7 s vs 71.2 s; text identical	No meaningful benefit observed on the tested MediaTek device; NNAPI path was removed.
Speaker identification	Two-speaker samples worked at clustering threshold 0.8; long noisy meeting produced up to 36 local clusters and took ~13 min	Long-meeting diarization was disabled/capped at recordings up to 20 minutes. Results should not be generalized to all recordings.
Model tool-call evaluation	71/71 tool names correct; 64/71 exact arguments; 1 false trigger in 150 chat prompts	Small test set only. Calculator exact arguments 12/16; set_timer 7/10. Not a production-readiness guarantee.